

50-mavzu. Ko'p o'zgaruvchili funktsiyaning differensial

50-ma'ruza

Reja

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1^o. Funktsiya differensial tushunchasi.

Faraz qilaylik, $f(x) = f(x_1, x_2, \dots, x_m)$ funktsiya $E \subset R^m$ da berilgan bo'lib, $x^0 = (x_1^0, x_2^0, \dots, x_m^0) \in E$ nuqtada differensiallanuvchi bo'lsin. Unda ta'rifga ko'ra funktsiyaning x^0 nuqtadagi to'liq orttirmasi

$$\Delta f(x^0) = \frac{\partial f(x^0)}{\partial x_1} \Delta x_1 + \frac{\partial f(x^0)}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f(x^0)}{\partial x_m} \Delta x_m + o(\rho) \quad (1)$$

bo'ladi. Bu munosabatda

$$\rho = \sqrt{\Delta x_1^2 + \Delta x_2^2 + \dots + \Delta x_m^2}$$

bo'lib, $\Delta x_1 \rightarrow 0, \Delta x_2 \rightarrow 0, \dots, \Delta x_m \rightarrow 0$ da $\rho \rightarrow 0$.

1-ta'rif. $f(x)$ funktsiyaning $\Delta f(x^0)$ orttirmasidagi

$$\frac{\partial f(x^0)}{\partial x_1} \Delta x_1 + \frac{\partial f(x^0)}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f(x^0)}{\partial x_m} \Delta x_m$$

ifoda $f(x)$ funktsiyaning x^0 nuqtadagi differensial (to'liq differensial) deyiladi va

$$df(x^0) \text{ yoki } df(x_1^0, x_2^0, \dots, x_m^0)$$

kabi belgilanadi:

$$df(x^0) = \frac{\partial f(x^0)}{\partial x_1} \Delta x_1 + \frac{\partial f(x^0)}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f(x^0)}{\partial x_m} \Delta x_m.$$

Demak, $f(x)$ funksiyaning x^0 nuqtadagi differensial $\Delta x_1, \Delta x_2, \dots, \Delta x_m$ larga bog'liq va ularning chiziqli funksiyasi bo'ladi.

Agar

$$\Delta x_1 = dx_1, \Delta x_2 = dx_2, \dots, \Delta x_m = dx_m$$

deyilsa, $f(x)$ funksiyaning x^0 nuqtadagi differensial ushbu

$$df(x^0) = \frac{\partial f(x^0)}{\partial x_1} dx_1 + \frac{\partial f(x^0)}{\partial x_2} dx_2 + \dots + \frac{\partial f(x^0)}{\partial x_m} dx_m \quad (2)$$

ko'rinishga keladi. Demak,

$$\Delta f(x^0) = df(x^0) + o(\rho).$$

Keyingi tenglikdan $\rho \rightarrow 0$ da

$$\Delta f(x^0) \approx df(x^0)$$

bo'lishi kelib chiqadi. Bu taqribiy formulaning mohiyati shundaki, funksiyaning orttirmasi $\Delta x_1, \Delta x_2, \dots, \Delta x_m$ larning, umuman aytganda murakkab funksiyasi bo'lgan holda funksiyaning differensial $\Delta x_1, \Delta x_2, \dots, \Delta x_m$ larning chiziqli funksiyai bo'lishidadir.

2^o. Murakkab funksiyaning differensial. Differensial shaklning invariantligi.

Aytaylik,

$$\frac{\partial f}{\partial t_k} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_k} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_k} + \dots + \frac{\partial f}{\partial x_m} \frac{\partial x_m}{\partial t_k}.$$

Bu xususiy hosilalarni (3) ifodadagi $\frac{\partial f}{\partial t_1}, \frac{\partial f}{\partial t_2}, \dots, \frac{\partial f}{\partial t_k}$ larning o'rniga qo'yamiz.

Natijada

$$\begin{aligned} df &= \left[\frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_1} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_1} + \dots + \frac{\partial f}{\partial x_m} \frac{\partial x_m}{\partial t_1} \right] dt_1 + \\ &+ \left[\frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_2} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_2} + \dots + \frac{\partial f}{\partial x_m} \frac{\partial x_m}{\partial t_2} \right] dt_2 + \\ &+ \dots + \left[\frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_k} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_k} + \dots + \frac{\partial f}{\partial x_m} \frac{\partial x_m}{\partial t_k} \right] dt_k = \\ &= \frac{\partial f}{\partial x_1} \left[\frac{\partial x_1}{\partial t_1} dt_1 + \frac{\partial x_1}{\partial t_2} dt_2 + \dots + \frac{\partial x_1}{\partial t_k} dt_k \right] + \\ &+ \frac{\partial f}{\partial x_2} \left[\frac{\partial x_2}{\partial t_1} dt_1 + \frac{\partial x_2}{\partial t_2} dt_2 + \dots + \frac{\partial x_2}{\partial t_k} dt_k \right] + \dots + \frac{\partial f}{\partial x_m} \left[\frac{\partial x_m}{\partial t_1} dt_1 + \frac{\partial x_m}{\partial t_2} dt_2 + \dots + \frac{\partial x_m}{\partial t_k} dt_k \right] \end{aligned}$$

bo'ladi.

Ravshanki,

$$\begin{aligned} \frac{\partial x_1}{\partial t_1} dt_1 + \frac{\partial x_1}{\partial t_2} dt_2 + \dots + \frac{\partial x_1}{\partial t_k} dt_k &= dx_1, \\ \frac{\partial x_2}{\partial t_1} dt_1 + \frac{\partial x_2}{\partial t_2} dt_2 + \dots + \frac{\partial x_2}{\partial t_k} dt_k &= dx_2, \\ &\dots\dots\dots \\ \frac{\partial x_m}{\partial t_1} dt_1 + \frac{\partial x_m}{\partial t_2} dt_2 + \dots + \frac{\partial x_m}{\partial t_k} dt_k &= dx_k. \end{aligned}$$

Demak, murakkab funksiyaning differensiali

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_m} dx_m \quad (4)$$

bo'ladi.

Biz yuqorida $f(x)$ hamda $f(x(t))$ murakkab funksiyaning differensiallari uchun (2) va (4) ifodalarni topdik. Bu ifodalarni solishtirib ularning formasi (shakli, ko'rinishi) bir xil, ya'ni (2) va (4) formulalarda funksiyaning differensiali xususiy hosilalarni

mos differensiallarga ko'paytmalardan tuzilgan yig'indiga teng ekanligini payqaymiz. Bu xossa differensial shaklning **invariantligi** deyiladi.

Eslatma. $f(x)$ funksiya differensialining (2) ifodasidagi $dx_1, dx_2, \dots, dx_m$ lar mos ravishda $\Delta x_1, \Delta x_2, \dots, \Delta x_m$ lar bo'lsa, $f(x(t))$ funksiya differensialidagi $dx_1, dx_2, \dots, dx_m$ lar $t_1, t_2, \dots, t_k$ o'zgaruvchilarning funksiyalari bo'ladi. Demak, (2) va (4) formulalarning ko'rinishlarigina bir xil bo'ladi.

3^o. Sodda qoidalar. Aytaylik,

$$u = u(x_1, x_2, \dots, x_m), \quad v = v(x_1, x_2, \dots, x_m)$$

funksiyalari $E \subset R^m$ to'plamda berilgan bo'lib, $x^\circ = (x_1^\circ, x_2^\circ, \dots, x_m^\circ) \in E$ nuqtada differensiallanuvchi bo'lsin. U holda:

$$1) \quad d(u + v) = du + dv,$$

$$2) \quad d(u \cdot v) = vdu + u dv,$$

$$3) \quad d\left(\frac{u}{v}\right) = \frac{vdu - u dv}{v^2} \quad (v \neq 0)$$

bo'ladi.

Bu munosabatlardan birini, masalan, 3) ning isbotini keltiramiz.

◀ Aytaylik,

$$F = \frac{u}{v}$$

bo'lsin. Bu holda F funksiya u va v larga va u va v lar o'z navbatida $x_1, x_2, \dots, x_m$ o'zgaruvchilarga bog'liq bo'lib, murakkab funksiyaga ega bo'lamiz. Differensial shaklning invariantli xossasiga ko'ra

$$dF = \frac{\partial F}{\partial u} du + \frac{\partial F}{\partial v} dv$$

bo'ladi. Ravshanki,

$$\frac{\partial F}{\partial u} = \frac{1}{v}, \quad \frac{\partial F}{\partial v} = -\frac{u}{v^2}$$

Demak,

$$dF = \frac{1}{v} du - \frac{u}{v^2} dv = \frac{vdu - u dv}{v^2},$$

ya'ni

$$d\left(\frac{u}{v}\right) = \frac{vdu - u dv}{v^2}$$

bo'ladi. ►

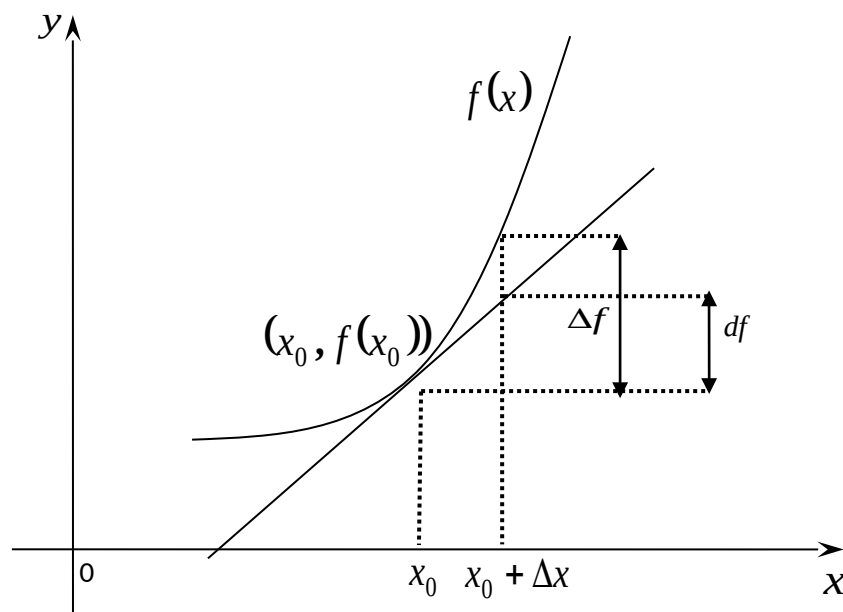
4⁰. Xususiylar. Funktsiya differensialining geometrik ma'nosi.

Aytaylik, $m=1$ bo'lsin. Bu holda $u=f(x)$ ($x \in R, u \in R$) funktsiya va uning differensial

$$du = df = f'(x)dx$$

ga ega bo'lamiz.

Ma'lumki, $u=f(x)$ funktsiyaning differensial shu funktsiya tasvirlangan egri chiziqqa $(x_0, f(x_0))$ nuqtada o'tkazilgan urinmaning ordinatasining orttirmasini ifodalaydi (27-chizma):



27-chizma

$m=2$ bo'lsin. Bu holda ikki o'zgaruvchili $u = f(x, y) \quad ((x, y) \in R^2, u \in R)$ funksiyaga ega bo'lib, uning (x_0, y_0) nuqtadagi differensial

$$du = df(x_0, y_0) = \frac{\partial f(x_0, y_0)}{\partial x} dx + \frac{\partial f(x_0, y_0)}{\partial y} dy \quad (5)$$

bo'ladi, bunda $dx = \Delta x$, $dy = \Delta y$.

Δx va Δy lar yetarlicha kichik bo'lganda

$$\Delta f(x_0, y_0) \approx df(x_0, y_0)$$

ya'ni

$$f(x_0 + \Delta x, y_0 + \Delta y) \approx f(x_0, y_0) + \frac{\partial f(x_0, y_0)}{\partial x_1} \Delta x + \frac{\partial f(x_0, y_0)}{\partial y} \Delta y$$

taqribiy formula hosil bo'ladi.

1-misol. Ushbu

$$u = x^y$$

funksiyaning differensialini topilsin.

◀ Ravshanki,

$$\frac{\partial u}{\partial x} = yx^{y-1}, \quad \frac{\partial u}{\partial y} = x^y \ln x.$$

Unda (5) formulaga ko'ra

$$du = yx^{y-1} dx + x^y \ln x dy$$

bo'ladi. ▶

2-misol. Tomonlari $x = 6m$ va $y = 8m$ bo'lgan to'g'ri to'rtburchak berilgan. Agar bu to'g'ri to'rtburchakning x tomonini 5 sm. ga oshirilsa, u tomonini 10 sm. ga kamaytirilsa, to'rtburchakning diagonalini qanchaga o'zgaradi?

◀ Agar berilgan to'g'ri to'rtburchakning diagonalini u desak, unda

$$u = \sqrt{x^2 + y^2}$$

bo'ladi. Endi

$$\Delta u(x_0, y_0) \approx \frac{x_0}{\sqrt{x_0^2 + y_0^2}} \Delta x + \frac{y_0}{\sqrt{x_0^2 + y_0^2}} \Delta y$$

bo'lishini e'tiborga olib, topamiz:

$$\Delta u(x_0, y_0) \approx \frac{x_0}{\sqrt{x_0^2 + y_0^2}} \cdot \Delta x + \frac{y_0}{\sqrt{x_0^2 + y_0^2}} \cdot \Delta y = \frac{x_0 \cdot \Delta x + y_0 \cdot \Delta y}{\sqrt{x_0^2 + y_0^2}}$$

Bu munosabatda

$$x_0 = 6 \text{ m}, \Delta x = 0,05 \text{ m}, y_0 = 8 \text{ m}, \Delta y = -0,10 \text{ m}$$

deyilsa, unda

$$\Delta u \approx \frac{6 \cdot 0,06 + 8 \cdot (-0,10)}{\sqrt{36 + 64}} \text{ m} = -0,05 \text{ m}$$

bo'lishi kelib chiqadi.

Demak, to'g'ri to'rtburchakning diagonalini taxminan 5 sm. ga kamaytirish mumkin. ►

Endi $f(x, y)$ funksiyasi differensialining geometrik ma'nosini keltiramiz.

Aytaylik,

$$z = f(x, y)$$

funksiya ochiq $E \subset R^2$ to'plamda differensiallanuvchi bo'lsin. Bu funksiya grafigi

R^3 fazoda biror $\Gamma(f)$ sirti ifodalasin. $\Gamma(f) = \{(x, y, z) \in R^3 : (x, y) \in E, z = f(x, y)\}$ sirtida

$(x_0, y_0, z_0) \in \Gamma(f)$ ($z_0 = f(x_0, y_0)$) nuqtani va shu nuqtadan o'tuvchi, qaralayotgan sirtga tegishli bo'lgan silliq

$$\Gamma = \{x = x(t), y = y(t), z = z(t) : \alpha \leq t \leq \beta\}$$

egri chiziqni olamiz. Modomiki, egri chiziq sirtida yotar ekan, unda

$$z(t) = f(x(t), y(t))$$

$$((x(t_0), y(t_0), z(t_0)) = (x_0, y_0, z_0), t_0 \in (\alpha, \beta))$$

bo'ladi. Ravshanki,

$$z(t) = f(x(t), y(t))$$

murakkab funksiya bo'lib, uning t_0 nuqtadagi differensial, differensial shaklning invariantligi xossasiga binoan, ushbu

$$df(x_0, y_0) = dz = \frac{\partial f(x_0, y_0)}{\partial x} dx + \frac{\partial f(x_0, y_0)}{\partial y} dy \quad (6)$$

ko'rinishga ega.

Koordinatalari dx, dy, dz bo'lgan vektor Γ egri chiziqqa (x_0, y_0, z_0) nuqtada o'tkazilgan urinma vektor bo'ladi.

Endi koordinatalari

$$-\frac{\partial f(x_0, y_0)}{\partial x}, -\frac{\partial f(x_0, y_0)}{\partial y}, 1$$

bo'lgan $\vec{n}$ vektorni qaraylik. Yuqoridagi (6) munosabat $\vec{n}$ vektor urinma vektorga (x_0, y_0, z_0) nuqtada ortogonal bo'lishini bildiradi. Shuning uchun $\vec{n}$ vektor egri chiziqqa (x_0, y_0, z_0) nuqtada ortogonal deyiladi.

Ma'lumki, Γ egri chiziq (x_0, y_0, z_0) nuqtadan o'tuvchi va $\Gamma(f)$ sirtida yotuvchi ixtiyoriy egri chiziq edi. Binobarin, $\vec{n}$ vektor shu (x_0, y_0, z_0) nuqtadan o'tuvchi va $\Gamma(f)$ sirtida yotuvchi ixtiyoriy egri chiziqqa ortogonal bo'ladi. Shuning uchun $\vec{n}$ vektor $\Gamma(f)$ sirtning nuqtasidagi normal vetori deyiladi.

Sirtning (x_0, y_0, z_0) nuqtasida o'tuvchi va sirtning normal vektoriga ortogonal bo'lgan tekislik, $\Gamma(f)$ sirtga (x_0, y_0, z_0) nuqtada o'tkazilgan urinma tekislik deyiladi. Uning tenglamasi

$$Z - z_0 = \frac{\partial f(x_0, y_0)}{\partial x}(X - x_0) + \frac{\partial f(x_0, y_0)}{\partial y}(Y - y_0)$$

bo'ladi, bunda (X, Y, Z) urinma tekislikdagi o'zgaruvchi nuqta. Bu tenglikdan foydalanib,

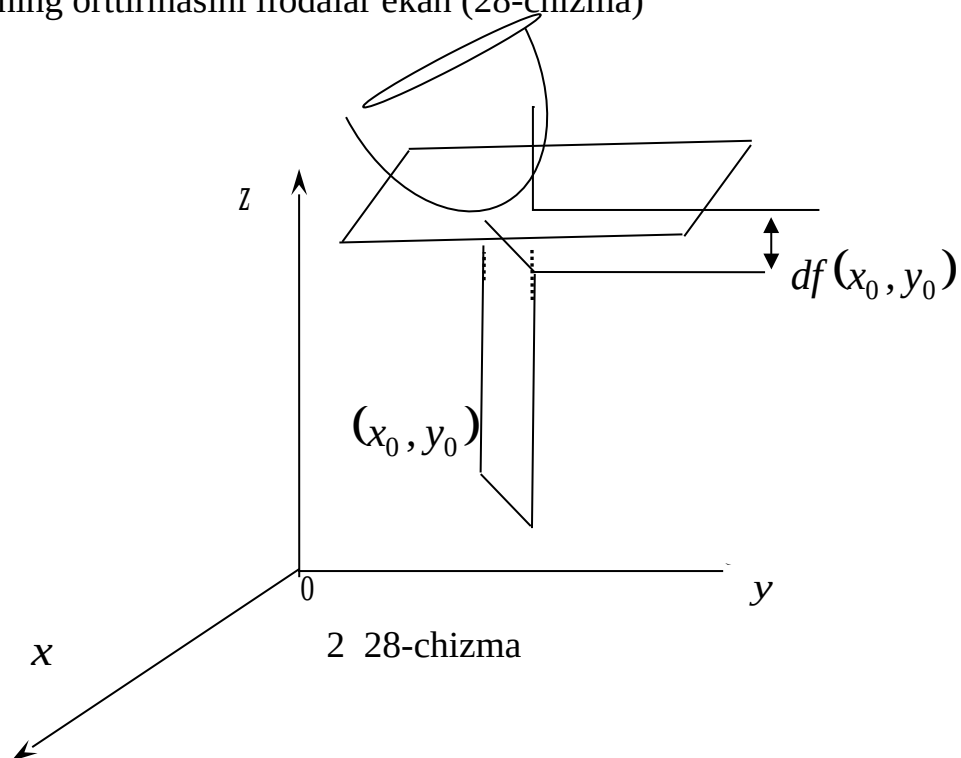
$$Z - z_0 = \frac{\partial f(x_0, y_0)}{\partial x}(x - x_0) + \frac{\partial f(x_0, y_0)}{\partial y}(y - y_0)$$

bo'lishini olamiz. Keltirilgan tenglik va (6) munosabatdan

$$df(x_0, y_0) = Z - z_0$$

bo'lishi kelib chiqadi.

Shunday qilib, $z = f(x, y)$ funksiyaning (x_0, y_0) nuqtadagi differensial $df(x_0, y_0)$ bu funksiya grafigiga $(x_0, y_0, f(x_0, y_0))$ nuqtasida urinma tekislik applikatsiyaning orttirmasini ifodalaydi (28-chizma)



28-chizma

Mashqlar

1. Ushbu

$$f\left(x^2 + y^2, \arctg \frac{y}{x}\right) \quad (x^2 + y^2 > 0)$$

funksiyaning differensialini topilsin.

2. Ushbu

$$\alpha = (1,02)^{3,01}$$

miqdorning taqribiy qiymati topilsin.

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